

4.

Exercício 4.1

a)

$f[x_, y_] = \{x, y\};$

$Jf(x, y) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

b)

$f[x_, y_] = \{x \text{Exp}[y] + \text{Cos}[y], x, x + \text{Exp}[y]\};$

$Jf(x, y) = \begin{pmatrix} e^y & e^y x - \text{Sin}[y] \\ 1 & 0 \\ 1 & e^y \end{pmatrix}$

c)

$f[x_, y_] = \{x y \text{Exp}[x y], x \text{Sin}[y], 5 x y^2\};$

$Jf(x, y) = \begin{pmatrix} e^{x y} y + e^{x y} x y^2 & e^{x y} x + e^{x y} x^2 y \\ \text{Sin}[y] & x \text{Cos}[y] \\ 5 y^2 & 10 x y \end{pmatrix}$

d)

$f[x_, y_, z_] = \{x - y, y + z\};$

$Jf(x, y, z) = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

e)

$f[x_, y_, z_] = \{x + y + \text{Exp}[z], x^2 y\};$

$Jf(x, y, z) = \begin{pmatrix} 1 & 1 & e^z \\ 2 x y & x^2 & 0 \end{pmatrix}$

Exercício 4.2

$f[x_, y_, z_] = \{x - y + z, x^2 y z, x y z\};$

$g[x_, y_, z_] = \{x y, y z, 2 x, x y z\};$

a)

$Df((-1, 0, -1); (2, 3, -1)) = \{-2, -3, 3\}$

$Dg((-1, 0, -1); (2, 3, -1)) = \{-3, -3, 4, 3\}$

b)

$Df(-1, 0, -1)(x, y, z) = \{x - y + z, -y, y\}$

$Dg(-1, 0, -1)(x, y, z) = \{-y, -y, 2 x, y\}$

Exercício 4.3

$f[x_, y_] = \{3 x, x + 2 y\};$

a)

$Jf(x, y) = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$

b)

A função f é de classe C^1 .

c)

$Df(1, 2)(x, y) = \{3 x, x + 2 y\}$

d)

$$Df(x_0,y_0)(x,y) = \{3x, x+2y\}$$

Exercício 4.4

$$f(x_, y_) = \{2x^2, 3y, 2xy\};$$

a)

$$Jf(x,y) = \begin{pmatrix} 4x & 0 \\ 0 & 3 \\ 2y & 2x \end{pmatrix}$$

b)

A função f é de classe C^1 .

$$Df(1,1)(x,y) = \{4x, 3y, 2x+2y\}$$

c)

$$Df(1,1)(2,3) = \{8, 9, 10\}$$

Exercício 4.5

a)

$$f(x_, y_) = 2xy / (x^2 + y^2)^2;$$

$$Hess f(x,y) = \begin{pmatrix} \frac{24y(x^3-xy^2)}{(x^2+y^2)^4} & -\frac{6(x^4-6x^2y^2+y^4)}{(x^2+y^2)^4} \\ -\frac{6(x^4-6x^2y^2+y^4)}{(x^2+y^2)^4} & \frac{24xy(-x^2+y^2)}{(x^2+y^2)^4} \end{pmatrix}$$

b)

$$f(x_, y_) = \cos[xy^2];$$

$$Hess f(x,y) = \begin{pmatrix} -y^4 \cos[xy^2] & -2y(xy^2 \cos[xy^2] + \sin[xy^2]) \\ -2y(xy^2 \cos[xy^2] + \sin[xy^2]) & -2x(2xy^2 \cos[xy^2] + \sin[xy^2]) \end{pmatrix}$$

c)

$$f(x_, y_) = \exp[-xy^2] + y^3x^4;$$

$$Hess f(x,y) = \begin{pmatrix} y^3(12x^2 + e^{-xy^2}y) & 2e^{-xy^2}y(-1 + 6e^{xy^2}x^3y + xy^2) \\ 2e^{-xy^2}y(-1 + 6e^{xy^2}x^3y + xy^2) & 2e^{-xy^2}x(-1 + 3e^{xy^2}x^3y + 2xy^2) \end{pmatrix}$$

d)

$$f(x_, y_) = 1 / (\cos[x]^2 + \exp[-y]);$$

$$Hess f(x,y) = \begin{pmatrix} \frac{2e^{2y}(e^y \cos[x]^4 - \sin[x]^2 + \cos[x]^2(1+3e^y \sin[x]^2))}{(1+e^y \cos[x]^2)^3} & \frac{4e^{2y} \cos[x] \sin[x]}{(1+e^y \cos[x]^2)^3} \\ \frac{4e^{2y} \cos[x] \sin[x]}{(1+e^y \cos[x]^2)^3} & -\frac{e^y(-1+e^y \cos[x]^2)}{(1+e^y \cos[x]^2)^3} \end{pmatrix}$$

Exercício 4.6

$$g(x_, t_) = 2 + \exp[-t] \sin[x];$$

$$\frac{\partial g}{\partial t}(x,t) = -e^{-t} \sin[x]$$

$$\frac{\partial^2 g}{\partial x^2}(x,t) = -e^{-t} \sin[x]$$

Exercício 4.7

$$f(x, y, z, w) = \exp[xyz] \sin[xw];$$

$$f_{xzw} = e^{xyz}xy((2+xyz)\cos[w] - wx\sin[w])$$

$f_{zwx} = e^{xyz} \cdot x \cdot y \cdot ((2 + x \cdot y \cdot z) \cdot \cos [w \cdot x] - w \cdot x \cdot \sin [w \cdot x])$

Exercício 4.8

a)

$D[2x^3, y]$

0

$D[x^2y + x, x]$

$1 + 2xy$

A função f seria de classe C^2 mas $f_{xy} \neq f_{yx}$

b)

$D[x \sin [y], y]$

$x \cos [y]$

$D[y \sin [x], x]$

$y \cos [x]$

A função f seria de classe C^2 mas $f_{xy} \neq f_{yx}$

Exercício 4.9

$f[0, 0] = 0;$
 $f[x_, y_] = x y^3 / (x^2 + y^2);$

a)

$f_x[0, 0] = 0$

$f_x[x, y] = \frac{y^3(-x^2 + y^2)}{(x^2 + y^2)^2}, \quad (x, y) \neq (0, 0)$

$f_y[0, 0] = 0$

$f_y[x, y] = \frac{x(3x^2y^2 + y^4)}{(x^2 + y^2)^2}, \quad (x, y) \neq (0, 0)$

b)

$f_{xy}[0, 0] = 1$

$f_{yx}[0, 0] = 0$

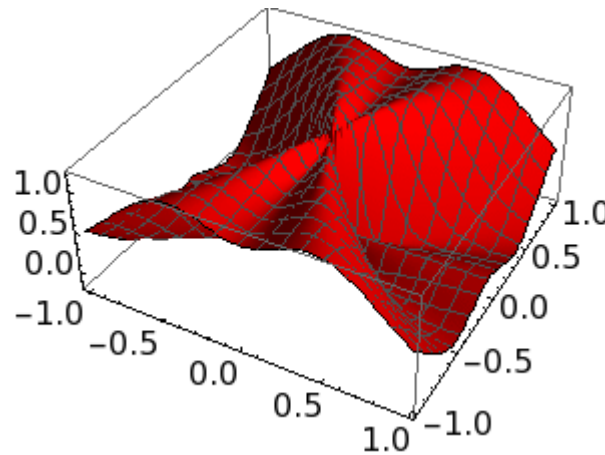
c)

$f_{xy}[x, y] = \frac{-3x^4y^2 + 6x^2y^4 + y^6}{(x^2 + y^2)^3}, \quad (x, y) \neq (0, 0)$

$Limit[f_{xy}[x, x], x \rightarrow 0]$

$\frac{1}{2}$

Conclui-se assim que f não é de classe C^2 .



Exercício 4.10


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u[x_, y_] = x y;  
v[x_, y_] = Sin[x y];  
w[x_, y_] = Exp[x];  
f[x_, y_, z_] = x^2 y + y^2 z;  
h[x_, y_] = f[u[x, y], v[x, y], w[x, y]]  
  
Grad[h[x, y], {x, y}] // Simplify  
  
{x^2 y^3 Cos[x y] + Sin[x y] (2 x y^2 + 2 e^x y Cos[x y] + e^x Sin[x y]),  
 x (x^2 y^2 Cos[x y] + 2 x y Sin[x y] + e^x Sin[2 x y])}
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Exercício 4.11

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f[x_, y_, z_] = x^2 y - x z;  
h[t_] = {a t^2, a t, t^3};
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a)

$Df(1,0,0)(1,2,2) = 0$

b)

$g'(t) = 5 a (-1 + a^2) t^4$
 $g'(t)=0 \Leftrightarrow a \in \{-1, 0, 1\}$

Exercício 4.13

a)

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u[x_, y_] = Log[Sin[x/y]]; x[t_] = 3 t^2;  
y[t_] = Sqrt[1 + t^2];
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$$\frac{du}{dt} = \left(-\frac{3 t^3}{(1 + t^2)^{3/2}} + \frac{6 t}{\sqrt{1 + t^2}} \right) \cot\left[\frac{3 t^2}{\sqrt{1 + t^2}}\right]$$

b)

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w[r_, s_] = r^2 + s^2; r[p_, q_] = p q^2; s[p_, q_] = p^2 Sin[q];
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$\frac{\partial w}{\partial p} = 2 p (q^4 + 2 p^2 \sin[q]^2)$

$\frac{\partial w}{\partial q} = 4 p^2 q^3 + p^4 \sin[2 q]$

c)

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z[x_, y_] = x^2 Sin[y]; x[s_, t_] = s^2 + t^2; y[s_, t_] = 2 s t;
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$\frac{\partial z}{\partial s} = 2 (s^2 + t^2) (t (s^2 + t^2) \cos[2 s t] + 2 s \sin[2 s t])$

$\frac{\partial z}{\partial t} = 2 (s^2 + t^2) (s (s^2 + t^2) \cos[2 s t] + 2 t \sin[2 s t])$